Ethan Wu

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EDUCATION

Northwestern University — **September 2026 – June 2030 (Anticipated)**
Computer Science & Operations Research (IEMS) — *Evanston, IL*

Commonwealth School — **September 2022 – June 2026**
High School — *Boston, MA*

SKILLS

Languages: Python, C++, Mojo, Rust, Go, JavaScript
Tooling & Frameworks: PyTorch, Google OR-Tools, Jax/Flax, LLVM, LightGBM, Optuna, Docker
Knowledge/Domains: Multi-Threading, Optimizations, Clustering Algorithms, ML, Git, Linux, SIMD, C-FFI

WORK EXPERIENCE

Research Intern — **October 2024 – Present**
MIT Jameel Clinic — *Cambridge, MA*

- Performed clustering analysis on public health data; **co-authoring 2 in-progress papers** targeting PLOS One.
- Developed novel method to find the best clustering from clusterings generated with different hyperparameters.
- Replicated & verified cancer prediction computational models in published papers.
- Co-designed **a custom featurizer** for a cancer prognostic machine learning model with an MIT PostDoc.
- Co-organized **a 60 student summer camp** about AI in healthcare in 2025 and 2026: *See LinkedIn Post*
- Built infrastructure software, such as a software deployment map and a summer program application portal.

PROJECTS

polars-mojo [Author] - Mojo, Apache Arrow, SIMD, Polars *https://prefix.dev/channels/modular-community/packages/polars-mojo*

- One of the first 60 libraries accepted into Modular's official Modular-Community index on prefix.dev
- Built zero-copy interop between Mojo and the Rust Polars dataframe library via Apache Arrow C Data Interface
- Implemented an RAII wrapper to synchronize Mojo `__deinit__` with Arrow C release callbacks to prevent leaks.
- Implemented SIMD reduction kernels (sum, mean, std) via Mojo's vectorize primitives.

Strata [Author] - Mojo, LAPACK/BLAS, SIMD *https://prefix.dev/channels/modular-community/packages/strata*

- One of the first 50 libraries accepted into Modular's official Modular-Community index on prefix.dev
- Developing a pure Mojo machine learning library with Sklearn ergonomics and zero Python runtime dependencies.
- Leveraged strict traits, static polymorphism, and generic SIMD types across float/int bit-widths to optimize ML models (Linear models, Random Forests, K-Means). Average of 1.9 times faster than Sklearn. *See benchmarks*
- Applied LAPACK/BLAS to handle linear algebra computation.

Kluster-Fudge (PyPI) [Author] - Python, JIT, GPU *https://pypi.org/project/kluster-fudge/*

- A Python library for clustering categorical data using K-Modes with various distance metrics (**3000+ downloads**).
- Implemented multi-threading, GPU acceleration, and JIT optimizations for better performance.

CubeCL (TracelAI) [Contributor] *https://github.com/tracel-ai/cubecl*

- Contributed upstream bug fix to CubeCL, a multi-platform GPU compute language extension for Rust.

Modular Community & Prefix Ecosystem (Modular AI) [Contributor] *https://github.com/modular/modular-community/pull/348*

- Resolved critical packaging automation failure by debugging and patching CI/CD scripts for the Mojo toolchain.

cox-hazards (Crates.io) [Author] - Rust *https://crates.io/crates/cox-hazards/*

- Built Rust library for survival analysis using Cox Proportional Hazards algorithm **(1000 downloads)**.
- Implemented Newton-Raphson and AdamW optimizers, with L1/L2 regularization for feature selection.

Kategorize (Crates.io) [Author] - Rust *https://crates.io/crates/kategorize*

- Built Rust library for K-Modes & K-Prototypes with various distance metrics (**950 downloads**).
- Multi-threaded and optimized with distance caching.